

Dense Expands, Sparse Anchors: Channel-Asymmetric Query Expansion for Hybrid Retrieval

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Abstract

LLM-based query expansion improves retrieval by generating document-like passages. In hybrid retrieval, however, most evaluations fuse fixed top- L dense and sparse rankings. Because the cutoff controls both which cross-channel contributions enter fusion and how much of each ranking is accessed, gains measured at one L can change or reverse at another. We separate these effects by evaluating retrieval effectiveness under complete-list fusion and recording the policy-specific per-channel replay stopping depths at which its ordered top- K is certified. We then introduce DESA (Dense Expansion and Sparse Anchoring), a channel-asymmetric query expansion method. An LLM generates complementary reference passages; orthogonal residual expansion adds their new semantic directions to the dense query, while score-product anchoring incorporates their lexical cues into sparse retrieval without broadening the original query’s lexical support. Across seven BEIR datasets, DESA improves nDCG@10 and Recall@20 over the unexpanded query by 3.82% and 2.38%, while reducing dense and sparse access depths by 36.90% and 36.56%. With equal dataset weighting, 63.31% of queries become shallower in both channels. However, both depths increase with Contriever on Touché-2020. These results support channel-specific integration of generated passages and joint evaluation of retrieval effectiveness and access depth.

1 Introduction

Existing LLM-based query expansion methods use generated content to construct retrieval queries or representations and typically evaluate retrieval gains at a preset top- L cutoff (Gao et al., 2023; Wang et al., 2023; Zhang et al., 2024; Liu and Zhang, 2025). In hybrid retrieval, however, top- L jointly determines how much of each ranking is read and which documents contribute to fusion; changing L may therefore change the fused result

(a) Dense expansion recovers missing semantics

Query A

How are various types of income taxed differently in the USA?

Generated references

Capital gains are taxed separately from **ordinary income**.
Long-term capital gains depend on the holding period.

Relevant passage

Rank 271 → 5

... **Long-term capital gains** are taxed below the marginal rate and differ from **wage or ordinary income** ...

Generated evidence supplies missing document language.

(b) Sparse anchoring preserves query evidence

Query B

How to find **cheaper alternatives** to a traditional **home telephone line**?

Generated references

Consider **VoIP services** as a home-phone alternative.
Some **internet service providers** offer integrated phone service.

Relevant passage

Rank 31 → 1

Use a **VOIP service provider** as an **alternative** to your **home phone line**.

Expansion-only match

Rank 5 → filtered

Internet Phone: VoIP dialer, reseller packages, login credentials, prices, and contact details ...

Green: generated | underline: original | Red: expansion-only

Figure 1: FiQA examples of DESA (Maia et al., 2018; Thakur et al., 2021). Dense expansion recovers missing document semantics; sparse anchoring promotes jointly supported passages and excludes expansion-only matches. Two references are shown per query; text is abridged and paraphrased, and ranks use draw 0.

itself (Zhang, 2026). A gain measured at a single cutoff is thus conditional on L and does not isolate the effect of query expansion.

To separate these factors, we first determine the fused result of each query construction from the complete dense and sparse rankings (Zhang, 2026). A fixed sound replay procedure then reveals the two rankings progressively and records the per-

channel depths at which the same ordered top- K is certified. This separates two questions: what fused result does an expansion produce, and how much ranked evidence is required to determine it? The answers need not move together. An expansion may improve retrieval effectiveness while pushing decisive evidence deeper into one channel. We therefore ask whether query expansion can improve effectiveness while reducing the access depth required from both channels.

Expansion effects are retriever-dependent (Weller et al., 2024; Li et al., 2026). On NFCorpus, applying the same expansion to both channels improves the fused result and reduces dense access depth, but increases sparse access depth. Generated terms substantially enlarge the set of documents with nonzero sparse scores, preventing the sparse ranking from exhausting as early as it does for the original query. This reversal motivates an asymmetric design: dense retrieval should absorb new semantic information, whereas sparse retrieval should restrict generated terms to documents already supported by the original query.

We address this asymmetry with DESA (Dense Expansion and Sparse Anchoring), a channel-asymmetric query expansion method for hybrid retrieval. Given an original query, the LLM generates several complementary reference passages around the same information need. For dense retrieval, orthogonal residual expansion preserves the original query direction and adds only the new semantic component supplied by the references. For sparse retrieval, score-product anchoring multiplies each document’s scores under the original and expanded queries. Documents matched only by generated terms receive no anchored score, while the new terms may still reorder documents supported by the original query. Together, these operations broaden dense semantic coverage without broadening sparse support: Dense expands; Sparse anchors. Figure 1 illustrates this division of labor through two retrieval examples.

Across seven BEIR datasets, DESA improves equal-dataset macro nDCG@10 and Recall@20 over the unexpanded query by 3.82% and 2.38%, while reducing dense and sparse access depths by 36.90% and 36.56%, respectively. With equal dataset weighting, 63.31% of queries become shallower in both channels. Mechanism experiments show that sparse anchoring contributes most of the effectiveness gain, while the dense orthogonal residual further reduces the ranked evidence

required to determine the fused result.

2 Related Work

2.1 LLM-Based Query Expansion

LLM-based query expansion supplements short queries with document-like language that may be absent from the original formulation. HyDE encodes a generated hypothetical document for dense retrieval, whereas Query2doc appends a pseudo-document to the query for sparse and dense retrieval (Gao et al., 2023; Wang et al., 2023). Later methods generate different forms of evidence or sample multiple references to cover alternative expressions of the same information need (Jagerman et al., 2023; Zhang et al., 2024). Although generated text can supply useful lexical and semantic cues, its effect varies across queries, datasets, and retrievers, and irrelevant content may reduce retrieval effectiveness (Weller et al., 2024).

2.2 Evaluating Expansion under Hybrid Fusion

Query-expansion methods commonly evaluate retrieval using a fixed top- L prefix from each ranking (Zhang et al., 2024; Liu and Zhang, 2025). In hybrid retrieval, L also determines which cross-channel contributions enter fusion, making the measured expansion effect conditional on the selected cutoff. Following EAHR, we evaluate complete-list retrieval effectiveness separately from channel access depth (Zhang, 2026), and ask whether query expansion can improve the former while reducing the latter.

2.3 Channel-Aware Integration

Generated content can enter retrieval through query text, dense representations, term weights, or rank fusion. Query2doc modifies the query text, MuGI pools representations from multiple references, Word2Passage reweights generated terms, and Exp4Fuse combines original and expanded rankings (Wang et al., 2023; Zhang et al., 2024; Choi et al., 2025; Liu and Zhang, 2025). DVCQR further constructs separate lexical and semantic views for sparse and dense retrieval (Li et al., 2026). These methods differ in where expansion is integrated; DESA instead focuses on how the same generated passages should be constrained differently across the two channels.

Recent work also combines heterogeneous retrievers per query: MoR weights sparse and dense

retrievers, while QuDAR adaptively fuses four rankings formed by original and expanded queries under both retriever types (Kalra et al., 2025; Kim et al., 2026). DESA instead modifies the dense and sparse rankings before fusion: generated passages expand dense semantic coverage while sparse retrieval remains anchored to the original query’s lexical support.

3 DESA

3.1 Method Overview

Given an original query q_o , DESA generates a set of complementary reference passages $R = \{r_1, \dots, r_n\}$. Dense retrieval keeps the original query as its base direction and adds only the new semantic component supplied by the references. Sparse retrieval combines the original and expanded scores so that generated terms can reorder documents already matched by the original query without introducing expansion-only matches. The resulting dense and sparse rankings are fused into the final result. Figure 2 summarizes this design: Dense expands; Sparse anchors.

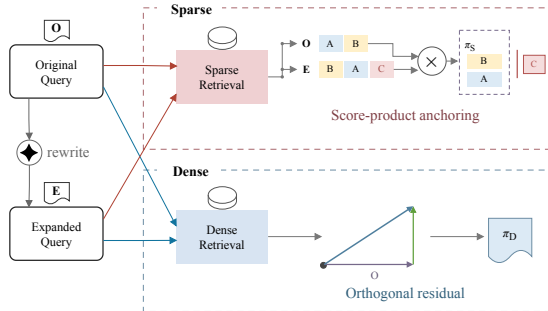


Figure 2: Design principle of DESA. The same generated passages expand dense semantic coverage but remain anchored to the original query’s lexical support in sparse retrieval.

3.2 Complementary Reference Generation

The generator is prompted to produce n short passages that could plausibly occur in relevant documents. Each passage introduces a different alias, technical term, or paraphrase while retaining the entities, relations, numbers, polarity, temporal conditions, scope, and other constraints of q_o . The prompt prohibits broader information needs and invented answers. The passages therefore vary in expression rather than intent, allowing the two retrieval channels to draw additional cues from a shared expansion. The complete prompt and decoding settings are provided in Appendix A.

3.3 Channel-Asymmetric Query Construction

3.3.1 Orthogonal Residual Expansion

Let f_D be the dense encoder. We encode the original query and each query–reference pair as unit vectors:

$$\begin{aligned} \mathbf{v}_o &= \text{norm}(f_D(q_o)), \\ \mathbf{v}_i &= \text{norm}(f_D(\text{concat}(q_o, r_i))), \\ \bar{\mathbf{v}}_r &= \frac{1}{n} \sum_{i=1}^n \mathbf{v}_i. \end{aligned}$$

The component of $\bar{\mathbf{v}}_r$ parallel to the original query is removed, and the remaining residual is added to $\mathbf{v}_o$:

$$\mathbf{z} = \bar{\mathbf{v}}_r - (\mathbf{v}_o^\top \bar{\mathbf{v}}_r) \mathbf{v}_o, \quad \mathbf{e}_D = \text{norm}(\mathbf{v}_o + \mathbf{z}).$$

Documents are ranked by their similarity to $\mathbf{e}_D$, producing the dense ranking π_D . Because $\mathbf{z} \perp \mathbf{v}_o$ and $\|\mathbf{z}\| \leq 1$,

$$\angle(\mathbf{e}_D, \mathbf{v}_o) = \arctan \|\mathbf{z}\| \leq \frac{\pi}{4}.$$

The references can therefore add semantic directions absent from the original representation while keeping the expanded query within 45° of it. If they contribute no orthogonal component, then $\mathbf{e}_D = \mathbf{v}_o$.

3.3.2 Score-Product Anchoring

For sparse retrieval, the expanded query concatenates the original query and all reference passages:

$$q_r = \text{concat}(q_o, r_1, \dots, r_n).$$

Let $s_S(q, d)$ be a non-negative additive sparse scorer. Each document d receives an original score, an expanded score, and their product:

$$\begin{aligned} s_o(d) &= s_S(q_o, d), & s_r(d) &= s_S(q_r, d), \\ s_A(d) &= s_o(d) s_r(d). \end{aligned}$$

Because q_r contains q_o , every document supported by the original query is also supported by the expanded query. Hence,

$$\text{supp}(s_A) = \text{supp}(s_o) \cap \text{supp}(s_r) = \text{supp}(s_o).$$

Sorting the positive anchored scores produces the sparse ranking π_S . Generated terms may promote or demote documents within the original support, but documents matched only by the expansion

receive zero anchored score and are excluded. If s_r is merely a positive multiple of s_o , the original sparse order is unchanged; positive global score rescaling also leaves the ranking invariant.

3.4 Complete Fusion and Access Measurement

We use the complete-list target and bound-based replay of Zhang (2026); DESA changes the query construction, not this evaluation protocol. Complete dense and sparse rankings are fused by equal-weight RRF (Cormack et al., 2009):

$$F(d) = \sum_{c \in \{D, S\}} \frac{1}{\kappa + \text{rank}_{\pi_c}(d)}.$$

A document absent from a channel contributes zero. Sorting by $F(d)$, with document identifiers breaking ties, defines the ordered target T_K . Retrieval effectiveness is evaluated directly on this complete-list result.

To measure channel access, a fixed replay procedure progressively reveals π_D and π_S . After L_c items have been read from channel c , the largest possible contribution from its next unread item is

$$u_c(L_c) = \frac{1}{\kappa + L_c + 1}.$$

The replay reads from the channel with the larger bound, breaking ties in favor of the dense channel, until unread documents can no longer change the membership or order of T_K . It records the replay stopping depths (L_D, L_S) required to determine the complete-list result.

4 Experiments

We first examine whether a fixed top- L cutoff changes the measured effect of query expansion. We then evaluate whether DESA improves retrieval effectiveness while reducing the ranking depth required from dense and sparse retrieval. The remaining experiments separate the contributions of the two channel operators and test robustness across generators, dense encoders, reference counts, and corpus sizes.

4.1 Evaluation Protocol

To measure cutoff sensitivity, we vary $L \in \{10, 20, 50, 100, 200, 500, 1000\}$. At each cutoff, the top- L dense and sparse rankings are fused using equal-weight RRF (Cormack et al., 2009). We compute nDCG@10 (Järvelin and Kekäläinen, 2002)

and Recall@20, compare the ordering of query constructions across cutoff depths, and measure whether each fixed- L result reproduces the ordered top-20 from complete-list fusion.

We also examine whether the cutoff changes the judgment of an expansion relative to the unexpanded query. For each query and generation draw, the metric difference between an expansion and Original is labeled as a gain, tie, or loss under both fixed- L and complete-list fusion. We report how often this label changes and separately count strict reversals, where a gain becomes a loss or vice versa.

All remaining experiments use the complete-list fusion defined in Section 3.4. Retrieval effectiveness is measured by nDCG@10 and Recall@20. Channel access is measured by the dense and sparse replay stopping depths (L_D, L_S) . We report their changes from Original and the proportion of queries for which both depths decrease. Sparse support size and exhaustion rate are used to explain changes in lexical access.

4.2 Datasets and Retrieval Setup

We evaluate on seven English BEIR datasets (Thakur et al., 2021): SciFact (Wadden et al., 2020), NFCorpus (Boteva et al., 2016), TREC-COVID (Voorhees et al., 2020), FiQA (Maia et al., 2018), ArguAna (Wachsmuth et al., 2018), Touché-2020 (Bondarenko et al., 2020), and SCIDOCS (Cohan et al., 2020). These datasets cover scientific claim verification, biomedical and financial retrieval, argument retrieval, and scientific document recommendation. Unless otherwise stated, the cutoff, controlled, and mechanism analyses use all seven datasets, with each dataset receiving equal weight in macro averages.

For the scale analysis, we construct nested TREC-COVID corpora containing 25,000, 50,000, 100,000, and all documents. Every snapshot retains the judged relevant documents, while the remaining documents are added in a fixed order. Each smaller corpus is therefore a strict subset of the larger ones.

The primary generator is Qwen2.5-7B-Instruct (Qwen Team, 2024). For each query, we perform three generation draws and produce five complementary reference passages per draw. The generator robustness experiment uses Mistral-7B-Instruct-v0.3 (Jiang et al., 2023) on up to 100 queries selected by a fixed hash from FiQA, ArguAna, Touché-2020, and SCIDOCS.

Dense retrieval uses BGE-small-en-v1.5 (Xiao et al., 2024), with normalized query and document vectors ranked by dot product. Sparse retrieval uses Lucene BM25 through Pyserini 2.3.0, with $k_1 = 0.9$ and $b = 0.4$ (Robertson and Zaragoza, 2009; Lin et al., 2021). Contriever (Izacard et al., 2022) is used as a second dense encoder. Score ties are broken by document identifier.

4.3 Comparison Methods

The controlled comparison includes Original and Shared expansion. Original submits the unexpanded query to both retrieval channels. Shared expansion uses the same generated references as DESA but applies them directly to both channels: dense retrieval averages the representations of the five query and reference pairs, while sparse retrieval concatenates the original query and all references. The comparison between Shared expansion and DESA therefore tests whether dense and sparse retrieval benefit from different integration rules.

We also compare with three representative generative expansion methods. HyDE encodes generated hypothetical documents for dense retrieval (Gao et al., 2023). Query2doc appends a generated pseudo-document to the query (Wang et al., 2023). MuGI generates multiple references, pools their contextualized dense representations, and uses query-weighted concatenation for sparse retrieval (Zhang et al., 2024). We evaluate all three reproduced pipelines on the same seven datasets as DESA.

We further compare with the rank-based QuDAR-simple variant, which uses the same generated references and fuses the top-1000 original-sparse (OS), expanded-sparse (ES), original-dense (OD), and expanded-dense (ED) rankings with RRF ($k = 60$) (Kim et al., 2026).

4.4 Mechanism and Robustness Analysis

We isolate the two channel operators with matched controls. For dense retrieval, the original sparse ranking is paired with either unprojected contextual expansion or orthogonal residual expansion. For sparse retrieval, the original dense ranking is paired with either direct rewriting or score-product anchoring. A 2×2 comparison further evaluates Original, each proposed operator alone, and their combination.

We then examine how each operator changes its retrieval channel. Dense behavior is measured by the angle between the original and expanded query

vectors. Sparse behavior is measured by top-20 turnover and by whether relevant documents move upward in the ranking. We compare these quantities with the corresponding changes in retrieval effectiveness and access depth using within-dataset quartiles and Spearman correlations.

Additional experiments vary the number of reference passages over $\{1, 3, 5\}$. Score-product anchoring is compared with a binary mask that retains expanded-query scores only for documents matched by the original query, separating support preservation from the use of graded original scores. We also test sparse retrieval using only the generated passages. Robustness experiments vary the RRF constant over $\{2, 20, 60, 100\}$, replace the generator and dense encoder, and evaluate the nested TREC-COVID corpora.

4.5 Statistical Analysis

The three generation draws are averaged within each query before aggregation. We report results for each dataset and equal-dataset macro averages. The 95% confidence intervals are obtained from 10,000 stratified bootstrap samples, with queries resampled within each dataset.

The primary comparisons are DESA against Original and DESA against Shared expansion. We use 10,000 stratified paired sign-flip randomizations and apply Holm correction across $n\text{DCG}@10$, $\text{Recall}@20$, L_D , and L_S separately for each comparator. The operator controls and the matched QuDAR comparison use the same draw-averaged, query-level paired procedure, with Holm correction across $n\text{DCG}@10$ and $\text{Recall}@20$.

5 Results

5.1 Fixed Cutoffs Alter Measured Effects

Figure 3 compares the query-level judgment of DESA relative to Original at $L = 50$ with the judgment under complete-list fusion. Each query-draw pair is labeled as a gain, tie, or loss from the corresponding metric difference. Diagonal cells preserve that judgment; off-diagonal cells show cases in which the fixed cutoff changes it.

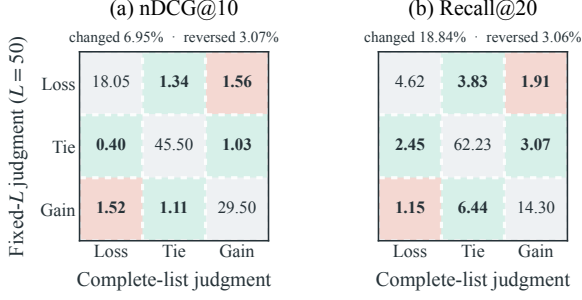


Figure 3: Transition between the DESA-versus-Original judgment at fixed $L = 50$ (rows) and under complete-list fusion (columns). Cells report the equal-dataset percentage of query-draw pairs. Green off-diagonal cells change among gain, tie, and loss; red cells are strict gain-loss reversals.

The fixed cutoff changes 6.95% of nDCG judgments and 18.84% of Recall judgments. Strict gain-loss reversals account for 3.07% and 3.06%, respectively. Thus, a practical fixed cutoff can change whether expansion is judged beneficial for an individual query, rather than merely perturbing the ranking. Results for other cutoffs and ordered top-20 agreement are reported in Appendix Table 5.

5.2 Retrieval Effectiveness and Access Depth

Table 1 summarizes the controlled comparison. Under complete-list fusion, DESA obtains the highest macro nDCG@10 and Recall@20 in the seven-dataset controlled comparison, reaching .4747 and .4695. Relative to Original, these values correspond to gains of 3.82% and 2.38%. The equal-dataset averages of the within-dataset replay stopping-depth reductions are 36.90% and 36.56% for the dense and sparse channels. The mean proportion of queries that become shallower in both channels is 63.31%. All four improvements remain significant after Holm correction ($p_{\text{Holm}} = .0004$).

Method	nDCG@10 ↑	Recall@20 ↑	L_D ↓	L_S ↓
<i>Original query</i>				
Original	0.4572	0.4586	1752.4	1455.1
<i>Shared expansion</i>				
Shared	0.4671	0.4666	607.4	606.7
<i>Single-channel construction</i>				
Dense only	0.4621	0.4637	1386.5	1219.9
Sparse only	<u>0.4714</u>	<u>0.4678</u>	884.4	822.0
<i>Channel-asymmetric construction</i>				
DESA	0.4747	0.4695	<u>728.9</u>	<u>673.2</u>

Table 1: Equal-dataset macro results for the seven-dataset controlled comparison. Higher is better for retrieval metrics; lower is better for replay stopping depths. Raw depths are averaged across datasets; percentage reductions in the text are computed within each dataset before macro-averaging.

Compared with Shared expansion, DESA improves macro nDCG by 1.61% and Recall by 0.61%. The absolute nDCG difference is +.0075 and remains significant after correction ($p_{\text{Holm}} = .0012$). The Recall difference is +.0028 ($p_{\text{Holm}} = .2736$), while the differences in dense and sparse depth are also nonsignificant. DESA therefore provides a significant nDCG advantage over Shared expansion.

NFCorpus illustrates the effect of sparse anchoring. Shared expansion increases the mean sparse support from 561.5 to 3211.6 documents and raises the sparse stopping depth by 105.81% relative to Original. DESA preserves the original support size and reduces sparse depth by 23.15%, while also reducing dense depth by 15.03%. Anchoring allows generated terms to improve the ordering of originally matched documents without expanding the lexical support.

5.3 Contributions of the Channel Operators

The 2×2 experiment shows that the two channel operators are complementary. Dense residual expansion alone improves nDCG and Recall by 1.08% and 1.11%, with dense and sparse depth reductions of 8.16% and 7.49%. Sparse anchoring alone produces larger effectiveness gains of 3.11% and 2.01%, together with depth reductions of 28.77% and 27.74%. Combining both operators gives the full DESA gains of 3.82% and 2.38%, while reducing dense and sparse depth by 36.90% and 36.56%.

Sparse anchoring contributes most of the effectiveness gain, while dense residual expansion further reduces access depth. Their isolated controls produce only small differences, whereas the complete channel-asymmetric construction significantly outperforms Shared expansion in nDCG. DESA’s advantage therefore comes from combining the two channel-specific operations.

Table 2 connects the two operators to their channel-specific outcomes on each dataset. DESA improves nDCG@10 over Original on all seven datasets. Adding sparse anchoring to Dense only improves all seven, while adding dense residual expansion to Sparse only improves six; FiQA is the sole exception at -0.66% . Under DESA, both replay stopping depths decrease on every dataset. Shared expansion broadens the positive-score sparse support by up to $5.72\times$, whereas DESA retains the Original support exactly on six datasets and by 99.9999% on Touché-2020, where

Dataset	Effectiveness: relative Δ nDCG@10 (%)			Access reduction vs. Original (%)		Sparse support ($\times$ Original)	
	DESA vs. Orig.	+Dense given Sparse	+Sparse given Dense	L_D	L_S	Shared	DESA
SciFact	+2.40	+1.12	+0.91	22.2	20.9	1.62 $\times$	1.00 $\times$
NFCorpus	+3.37	+2.14	+1.36	15.0	23.2	5.72$\times$	1.00 $\times$
TREC-COVID	+5.69	+0.51	+4.68	74.9	66.7	1.53 $\times$	1.00 $\times$
FiQA	+3.61	-0.66	+3.74	49.7	48.7	1.64 $\times$	1.00 $\times$
ArguAna	+1.27	+0.32	+0.88	15.2	15.3	1.00 $\times$	1.00 $\times$
Touché-2020	+7.52	+0.92	+5.70	45.8	45.8	1.88 $\times$	1.00 $\times$
SCIDOCs	+0.86	+0.03	+0.13	35.4	35.4	2.02 $\times$	1.00 $\times$
Pattern	7/7 improve	6/7 improve	7/7 improve	7/7 shallower	7/7 shallower	up to 5.72 $\times$	$\approx$ 1.00 $\times$

Table 2: Per-dataset evidence for channel-asymmetric construction. Effectiveness columns report relative nDCG@10 changes; +Dense and +Sparse compare DESA with the corresponding one-operator construction. Access columns report replay-depth reductions from Original, where positive values indicate shallower access. Sparse support is normalized by Original; red marks the sole degradation.

the exported tail omits 25 extremely low-scoring document occurrences.

The cross-dataset pattern makes the division of labor explicit: sparse anchoring supplies the more consistent effectiveness gain, while dense residual expansion adds complementary gains on most datasets. Together, they improve retrieval without broadening sparse support and reduce the ranked evidence required by both channels. Detailed angle, turnover, relevant-rank, and operator-control diagnostics are reported in Appendix B.2.

Additional binary-mask, reference-count, and operator-control results are reported in Appendix C.

5.4 Comparison with Recent and Prior Methods

Under matched generated references, QuDAR-simple RRF obtains 0.4761 nDCG and 0.4714 Recall, compared with 0.4747 and 0.4695 for DESA. Neither paired difference is significant after correction. After accounting for finite sparse lists, QuDAR-simple fuses a mean of 3,911.5 available entries across its four top-1000-capped rankings. DESA requires a mean of 1,402.1 entries to certify its fused result under the replay policy, a 64.16% reduction in equal-dataset mean entry count (Table 3).

Across all seven datasets, DESA’s macro nDCG and Recall exceed HyDE’s by 15.39% and 7.32%. Averaging percentage reductions within datasets, its dense and sparse stopping depths are 80.48% and 80.26% lower. Relative to MuGI, the macro effectiveness gains are 3.56% and 2.15%, and the

Method	nDCG@10 $\uparrow$	Recall@20 $\uparrow$	Entries $\downarrow$
<i>Matched references, seven datasets</i>			
QuDAR-simple	0.4761	0.4714	3911.5
DESA	<u>0.4747</u>	<u>0.4695</u>	1402.1
<i>Prior query expansion, seven datasets</i>			
Original	0.4572	0.4586	3207.6
HyDE	0.4114	0.4374	16668.8
MuGI	0.4584	0.4596	2790.1
Query2doc	<u>0.4728</u>	<u>0.4675</u>	<u>1771.5</u>
DESA	0.4747	0.4695	1402.1

Table 3: External comparisons over the same seven evaluation datasets. Rank entries are the available items in QuDAR-simple’s four top-1000-capped lists and $L_D + L_S$ for the two-channel methods.

corresponding depth reductions are 24.36% and 37.21%. These comparisons show higher mean effectiveness and lower mean access depths.

In the lower block, DESA has 0.40% higher nDCG and 0.43% higher Recall than Query2doc, with 11.85% and 23.28% lower dense and sparse stopping depths.

5.5 Robustness and Failure Boundary

DESA retains its effectiveness gains across the tested generator, dense encoder, and corpus sizes. Access depth decreases in most settings, with Contriever on Touché-2020 as the only observed failure case. Detailed generator, encoder, and corpus-scale results are reported in Appendix C.1.

6 Conclusion

This work shows that query expansion in hybrid retrieval should account for the distinct matching behavior of dense and sparse retrieval channels. DESA implements this channel-asymmetric

view using a shared set of generated reference passages. Orthogonal residual expansion adds complementary semantic directions to the dense query, whereas score-product anchoring uses the same generated passages to reorder documents within the original sparse support. To evaluate this design without tying its measured effect to a fixed top- L cutoff, we determine the fused ranking from complete channel lists and record the per-channel replay stopping depths needed to certify its ordered top- K .

Across seven BEIR datasets, DESA improves equal-dataset macro nDCG@10 and Recall@20 over Original by 3.82% and 2.38%, while reducing dense and sparse replay stopping depths by 36.90% and 36.56%. The operator analysis clarifies the source of these gains: sparse anchoring provides the more consistent effectiveness improvement, and dense residual expansion contributes additional gains while reducing the ranked evidence required for fusion. Under matched generated references, paired tests do not detect a statistically significant effectiveness difference between DESA and QuDAR-simple RRF. Comparing their equal-dataset mean entry counts, DESA uses 64.16% fewer available fusion-side rank entries.

Together, these findings point to a division of labor for query expansion in hybrid retrieval. Generated reference passages broaden semantic coverage in dense retrieval and, when anchored to the original query, improve ordering in sparse retrieval without broadening lexical support. This division of labor captures the central design principle of DESA: Dense expands, while Sparse anchors.

Limitations

Our evaluation covers seven English BEIR datasets, two generators, two dense encoders, BM25, and equal-weight RRF. The results may differ for other languages, learned sparse retrievers, or fusion rules. Score-product anchoring further assumes a non-negative sparse scoring function.

Replay stopping depth measures the ranked evidence needed to certify the fused top- K , rather than wall-clock latency. End-to-end efficiency also depends on generation, encoding, indexing, and the retrieval interface. Generated references may contain irrelevant or unsupported expressions, while pooled relevance judgments may omit relevant documents outside existing pools. Evaluating DESA with broader retrieval architectures and online in-

cremental retrieval would clarify these remaining boundaries.

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A Reproducibility Details

This section records the generation, query-construction, retrieval, and replay settings needed to reproduce the experiments.

A.1 Complementary Reference Generation

We use the following Qwen2.5-7B-Instruct revision for both the model and tokenizer:

fe11104b620d588ccc049ff6631dd3ea002e3d98

The unquantized BF16 model runs with MLX-LM 0.31.2. Decoding uses a temperature of 0.7, top- p of 0.9, top- k of 20, and a maximum of 256 new tokens. Queries are sorted by identifier and processed in batches of eight. Each batch seed is derived from the protocol version, dataset, prompt hash, and batch index using the first 64 bits of their SHA-256 digest.

Prompt 1. Complementary Reference Generation

System prompt

You generate evidence-like reference passages for information retrieval.

Given one search query, return exactly five concise, complementary passages that could plausibly occur in relevant documents. Preserve every named entity, relation, number, polarity, temporal condition, scope, and constraint in the query. Add useful aliases, technical terminology, and paraphrases, but do not broaden the information need or invent a specific answer. Each passage must be one sentence of at most 25 words.

Return one single-line JSON array containing exactly five distinct strings. Count the five array items before responding. Do not output an object key, query identifier, Markdown, placeholders, or any other text. Every item must contain a substantive passage.

User input

`{"query": "{query}"}`

The user input is serialized as a canonical JSON object with one query field. XGrammar 0.2.1 requires exactly five strings; validation rejects empty, duplicate, or placeholder passages. Invalid output is retried once with the same seed and validation feedback, then falls back to the unexpanded query.

The second generator is Mistral-7B-Instruct-v0.3 at revision

c170c708c41dac9275d15a8fff4eca08d52bab71.

It uses the same decoding, batching, and validation settings.

A.2 Query Construction

Orthogonal residual expansion uses float32 throughout. Shared expansion uses the same query [SEP] reference representations, but averages and normalizes them without removing the component parallel to the original query.

For sparse retrieval, the expanded query contains one copy of the original query followed by the selected reference passages. BM25 scores under the original and expanded queries are stored separately before applying the document-wise score product. The binary-mask ablation retains the expanded-query score only for documents matched by both queries; the references-only ablation removes the original query. DESA neither repeats the original query nor uses MuGI’s repetition coefficient.

A.3 Baseline Reproduction

All generative baselines use the same primary generator. Exact prompts and the reproduction code are available at <https://github.com/ln-one/hybrid-query-construction>.

HyDE generates eight hypothetical documents per run and averages their normalized representations; its sparse channel uses the original query. FiQA, ArguAna, SciFact, and TREC-COVID use task-specific instructions, the remaining datasets use a general instruction, and generation is capped at 512 new tokens.

Query2doc generates one pseudo-document of at most 120 words. Dense retrieval encodes it with the original query; sparse retrieval concatenates it with five copies of the original query.

MuGI generates five pseudo-references. Dense retrieval averages the five query–reference representations. Sparse retrieval repeats the original query

$$\left\lfloor \frac{\sum_{i=1}^n |r_i|}{|q_o|^\beta} \right\rfloor$$

times before appending all references, where $|\cdot|$ denotes character length and $\beta = 4$. We reproduce MuGI’s generation and query integration components without its pseudo-relevance feedback calibration.

A.4 Retrieval Implementation

BGE-small-en-v1.5 uses revision

5c38ec7c405ec4b44b94cc5a9bb96e735b38267a.

Queries are prefixed with “Represent this sentence for searching relevant passages:” and encoded with a maximum length of 512 tokens. Contriever uses mean pooling at revision 2bd46a25019aeea091fd42d1f0fd4801675cf699.

All query and document embeddings are normalized. Pyserini 2.3.0 runs with Java 21 and uses DefaultEnglishAnalyzer. A separate Lucene

index is built for each dataset and each TREC-COVID snapshot, so BM25 collection statistics are computed within the corresponding corpus. Documents with equal retrieval or fusion scores are ordered by the UTF-8 order of their identifiers.

A.5 Access-Depth Replay

This implements the fixed bound-based replay protocol adopted in Section 3.4 from Zhang (2026). The replay starts with empty dense and sparse prefixes. After reading l_c items from channel c , its next unread contribution is bounded by

$$u_c(l_c) = \frac{1}{60 + l_c + 1}.$$

The bound becomes zero after exhaustion. The replay reads the channel with the larger bound, breaking ties toward dense. An observed document’s accumulated score is a lower bound; an unseen contribution is bounded by the corresponding next-unread value, and a document unseen in both channels by their sum. Replay stops when these bounds certify top-20 membership and order, recording $(L_D, L_S) = (l_D, l_S)$.

A.6 Artifact Traceability

Generation records store the raw and parsed output, hashes, decoding settings, seed, retries, runtime lock, and code commit. Query-level records store the method, generation run, effectiveness, (L_D, L_S) , sparse support and exhaustion, ordered top-20, and hashes of the compressed rankings, configuration, generation, and replay trace.

The Python environment is fixed by `uv.lock`; Java is checked before retrieval. A hash manifest records code, configurations, prompts, inputs, models, generations, and rankings. Reported tables are rebuilt from the query-level records.

B Operator Properties and Mechanism Diagnostics

B.1 Properties of the Channel Operators

For the dense construction, each contextual vector $\mathbf{v}_i$ is unit normalized. The triangle inequality gives

$$\|\bar{\mathbf{v}}_r\| \leq \frac{1}{n} \sum_{i=1}^n \|\mathbf{v}_i\| = 1.$$

Orthogonal projection cannot increase the norm, so $\|\mathbf{z}\| \leq 1$. Since $\mathbf{v}_o^\top \mathbf{z} = 0$ and $\|\mathbf{v}_o\| = 1$,

$$\mathbf{e}_D^\top \mathbf{v}_o = \frac{1}{\sqrt{1 + \|\mathbf{z}\|^2}},$$

$$\angle(\mathbf{e}_D, \mathbf{v}_o) = \arctan \|\mathbf{z}\| \leq \frac{\pi}{4}.$$

If $\mathbf{z} = \mathbf{0}$, normalization leaves $\mathbf{v}_o$ unchanged.

For sparse retrieval, non-negativity gives $\text{supp}(s_o s_r) = \text{supp}(s_o) \cap \text{supp}(s_r)$. Additivity and the inclusion of q_o in q_r imply $\text{supp}(s_o) \subseteq \text{supp}(s_r)$, yielding exact support preservation. If $s_r = c s_o$ for $c > 0$, then $s_A = c s_o^2$; squaring is strictly increasing on positive scores, so the ranking is unchanged. Multiplying s_o or s_r by a positive global constant only rescales every anchored score and likewise leaves the ranking unchanged.

B.2 Mechanism Diagnostics

Table 4 reports dense angle, one minus sparse top-20 overlap with Original, and the mean reciprocal-rank change of judged relevant documents present in the exported rankings. Macros weight datasets equally.

Figure 4 shows the query-level distributions behind the aggregate diagnostics. One invalid-generation fallback leaves 11,327 query–draw pairs. Their mean dense angle is 11.98° and the maximum is 27.87° , below the analytic 45° bound. Sparse anchoring changes 26.15% of the original sparse top-20 on average.

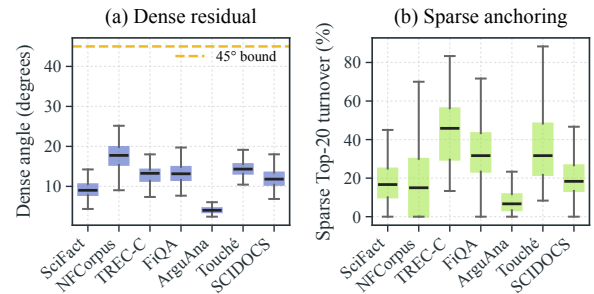


Figure 4: Realized behavior of the two channel operators over three generation draws on seven datasets. Dense angles are measured from the original query; sparse turnover is one minus top-20 overlap with the original sparse ranking. Boxes show the interquartile range and median; whiskers extend to 1.5 times the interquartile range, with outliers omitted.

Within-dataset quartiles show no monotonic relation between either diagnostic and effectiveness or access depth. Relevant-document reciprocal-rank gain correlates more strongly with sparse-only nDCG gain than overall turnover does (.391 versus .078); dense-angle correlations with nDCG gain

Dataset	Dense angle (°)	Sparse top-20 turnover (%)	Relevant RR change	Support retained (%)
SciFact	9.38	18.88	.0211	100.0000
NFCorpus	17.65	18.69	.0147	100.0000
TREC-COVID	12.99	45.37	.0021	100.0000
FiQA	13.34	33.91	.0204	100.0000
ArguAna	4.02	8.14	.0048	100.0000
Touché-2020	14.40	37.07	.0091	99.9999
SCIDOCS	12.05	20.98	.0028	100.0000
Equal-dataset macro	11.98	26.15	.0107	100.0000

Table 4: Mechanism diagnostics computed from the frozen query constructions and rankings. Support retained is the mean per-query–draw fraction of the original sparse support; the displayed macro rounds to 100.0000%.

and depth reduction are .010 and .011. The data therefore support no angle or turnover threshold.

Exact support preservation assumes a non-negative additive scorer and complete score export. It holds for six datasets. Touché-2020 omits 25 extremely low-scoring tail documents across 23 of 147 query–draw pairs; mean and minimum retention are 99.99990% and 99.99881%. We retain these cases and do not call the Lucene export exactly support preserving there.

C Additional Experimental Results

C.1 Additional Ablations and Robustness

Binary masking preserves sparse support but removes original-score magnitude, lowering macro nDCG@10 by 1.54% and Recall@20 by 0.74%. Increasing references from one to three improves nDCG@10 by 1.91% and reduces dense/sparse replay stopping depths by 17.23%/18.14%; using five adds 0.35% nDCG@10 and 0.43% Recall@20, with further reductions of 3.83%/3.40%.

With Mistral, all four sampled datasets retain non-negative effectiveness changes and 7.12–47.61%/7.16–47.62% dense/sparse depth reductions; 10% of sampled ArguAna queries fall back. Contriever improves both metrics on all four datasets but raises Touché-2020 depths by 14.29%/4.34%. Nested TREC-COVID gains and reductions remain positive but non-monotonic.

C.2 Matched Operator and QuDAR Controls

Tables 6 and 7 report matched controls. Operator pairs share references, keep the other channel at Original, and average three draws within query before inference.

Construction	nDCG@10	Recall@20
Original	.4572	.4586
Contextual Dense	.4622	.4637
Residual Dense	.4621	.4637
Rewrite Sparse	.4686	.4670
Product Sparse	.4714	.4678
Paired effect (proposed – control)		
Dense residual, nDCG	–.00005 [–.00086, .00082], $p_{\text{Holm}} = 1.0$	
Dense residual, Recall	–.00006 [–.00043, .00033], $p_{\text{Holm}} = 1.0$	
Sparse product, nDCG	.00283 [–.00126, .00693], $p_{\text{Holm}} = .3650$	
Sparse product, Recall	.00078 [–.00184, .00331], $p_{\text{Holm}} = .5689$	

Table 6: Macro operator controls and query-level paired effects. Holm correction covers nDCG and Recall within each control pair.

Dataset	nDCG@10		Recall@20	
	DESA	QuDAR	DESA	QuDAR
ArguAna	.3991	.3985	.9237	.9253
FiQA	.3780	.3833	.5450	.5554
NFCorpus	.3793	.3780	.2195	.2227
SCIDOCS	.1940	.1957	.2787	.2811
SciFact	.7427	.7403	.9200	.9211
TREC-COVID	.8224	.8233	.0435	.0438
Touché-2020	.4071	.4137	.3560	.3500
Equal-dataset macro	.4747	.4761	.4695	.4714

Table 7: Matched-reference comparison with QuDAR–simple RRF (top-1000 original-sparse (OS), expanded-sparse (ES), original-dense (OD), and expanded-dense (ED) rankings; $k = 60$). DESA–QuDAR paired effects are –.00145 [–.00478, .00181] for nDCG and –.00188 [–.00400, .00019] for Recall ($p_{\text{Holm}} = .3966$ and .1668).

(a) Per-dataset controlled results

Dataset	Method	Queries	nDCG@10	Recall@20	L_D	L_S	Sparse support	Exhaust. (%)
ArguAna	Original	1406	.3941	.9203	157.1	156.7	8587.8	0.00
	Shared		.4004	.9265	123.4	123.0	8616.8	0.00
	DESA		.3991	.9237	133.1	132.7	8587.8	0.00
FiQA	Original	648	.3649	.5260	2179.0	2076.3	28384.6	1.23
	Shared		.3635	.5391	1104.1	1103.6	46673.3	0.00
	DESA		.3780	.5450	1096.2	1065.6	28384.6	0.93
NFCorpus	Original	323	.3670	.2099	474.4	173.7	561.5	48.61
	Shared		.3780	.2260	359.1	357.6	3211.6	1.14
	DESA		.3793	.2195	403.1	133.5	561.5	46.75
SCIDOCS	Original	1000	.1923	.2721	694.8	599.3	11503.0	2.70
	Shared		.1905	.2790	372.7	372.2	23239.6	0.00
	DESA		.1940	.2787	448.8	387.2	11503.0	1.63
SciFact	Original	300	.7253	.9036	409.1	368.6	2765.7	4.33
	Shared		.7325	.9182	284.1	282.6	4489.0	0.33
	DESA		.7427	.9200	318.3	291.6	2765.7	3.89
TREC-COVID	Original	50	.7781	.0398	6270.9	4729.9	92683.8	2.00
	Shared		.8169	.0433	787.3	786.9	141483.1	0.00
	DESA		.8224	.0435	1574.8	1574.4	92683.8	0.00
Touché-2020	Original	49	.3786	.3382	2081.8	2081.4	151619.0	0.00
	Shared		.3883	.3344	1221.5	1221.3	284677.0	0.00
	DESA		.4071	.3560	1127.8	1127.4	151618.9	0.00

(b) Primary statistical tests

Comparison / outcome	Effect [95% CI]	p/p_{Holm}
Original / nDCG@10	.0175 [.0111, .0238]	.0001 / .0004
Original / Recall@20	.0109 [.0070, .0151]	.0001 / .0004
Original / L_D	1023.6 [377.5, 2012.6]	.0001 / .0004
Original / L_S	781.9 [356.1, 1339.7]	.0001 / .0004
Shared / nDCG@10	.0075 [.0036, .0116]	.0003 / .0012
Shared / Recall@20	.0028 [-.0003, .0061]	.0912 / .2736
Shared / L_D	-121.4 [-367.0, 39.7]	.3157 / .6313
Shared / L_S	-66.5 [-310.6, 93.0]	.7909 / .7909

(c) Fixed-cutoff diagnostics

L	Exact top-20 (%)	nDCG (%)		Recall (%)	
		Change	Reverse	Change	Reverse
20	1.92	18.23	8.84	22.50	3.84
50	16.54	6.95	3.07	18.84	3.06
100	47.40	2.17	0.73	11.74	0.99
1000	92.62	0.19	0.01	1.18	0.00

Table 5: Additional results. (a) Dataset-level controlled results. (b) Stratified paired tests after within-query averaging over three generation draws. (c) DESA query-draw sensitivity to fixed top- L , macro-averaged with equal dataset weight; exact top-20 compares with complete-list fusion, and the conclusion columns compare DESA with Original.